

MERRILL KEATING

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SUMMARY

Mechanical engineer with hands-on experience in aerospace systems, structural analysis, and test engineering. NASA Space Grant research on Mars entry trajectories and autonomous robotics. Background in FEA, CAD, GD&T, and design-build-test cycles from competition robotics to university rover design. Strong Python and MATLAB skills. Passionate about safe human spaceflight and extending human presence beyond Earth.

TECHNICAL SKILLS

CAD & Analysis: SolidWorks, AutoCAD, Inventor, FEA (structural, thermal, modal), GD&T (ASME Y14.5), tolerance analysis

Mechanical Systems: Mechanisms, drive trains, stepper motors, brushless DC motors, servos, actuators, power systems

Test & Instrumentation: Test fixture design, sensor integration, IMU data acquisition, test planning & execution, data analysis

Fabrication: 3D printing, CNC machining, mill, lathe, bandsaw, CO2 laser cutting, soldering, wiring

Programming: Python, MATLAB, C++, ROS

ENGINEERING EXPERIENCE

NASA Space Grant Research — AIRLAB | Research Fellow | University of Washington | 2021–2023

- Developed Python simulation modeling Mars entry, deceleration, and aerobraking for insect-sized robotics
- Performed FEA simulations for structural loads during reentry and shock absorption
- Designed self-resettable hopping robot mechanisms for extraterrestrial terrain traversal
- Tested brushless motors with current limiting to select optimal actuators; evaluated power requirements
- Presented findings at UW 25th Annual Undergraduate Research Symposium

Wave Energy Conversion Research (Capstone) | Lead Engineer | UW + National Lab | 2023–2024

- Led design, fabrication, and testing of spring-integrated pendulum system for wave energy conversion
- Built electromechanical test rig with microcontroller-driven actuators and embedded sensors
- Applied GD&T and tolerance analysis in 3D CAD to ensure robustness under dynamic loads
- Developed Python code for IMU data acquisition and signal processing
- Collaborated with national lab researchers on test planning and system integration

UW Husky Robotics — Chassis Subsystem | Team Member | 2020–2021

- Designed chassis and wheelbase for University Rover Challenge Mars-terrain simulation
- Worked with stepper motors, brushless DC motors, and servos for drive and articulation systems
- Integrated ROS for control systems and sensor communication
- Conducted materials research for wheel composition under simulated Martian conditions

FIRST Robotics Team 4915 | Mechanical Lead | 2018–2020

- Designed and built competition robots with drive trains, mechanisms, and microcontroller-based control
- Hands-on fabrication: CNC, mill, lathe, laser cutting, 3D printing, soldering, wiring harnesses
- Applied GD&T and tolerance analysis in CAD for precision mechanism design
- Awards: District Event Winner, Engineering Inspiration Award (2x), Industrial Design Award

Keating Consulting Services, Inc. | IT Consultant | 2019–Present

- Device deployment, network configuration, and system troubleshooting for K-12 districts
- Client-facing project management; database development and system migrations

EDUCATION

University of Washington — B.S. Mechanical Engineering | August 2025

Early entrance program (began at 15) • Resident Adviser 2021–2023

Relevant Coursework: FEA, Structures, Mechanics of Materials, Machine Design, Microcontroller Systems, Machining

ADDITIONAL

Leadership: Founded The Power of 100 Girls & Girls Ignited (226K+ community). Board Member, West Sound Technology Association. 10 years mentoring at West Sound CoderDojo.

Recognition: Grace Hopper Award (2025) • USA Today Women of the Year, WA (2024) • Outstanding Young Philanthropist (2024)

Publications: 5 books on AI ethics & human-machine collaboration (The Merrill Protocol series)